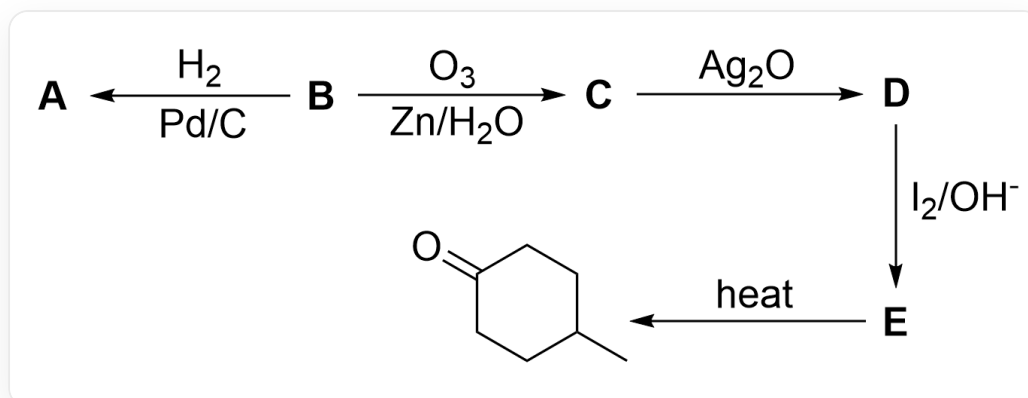


Question



This image describes several steps of organic synthesis. The substrate is **B** with an unknown structure, which reacts with H_2 , Pd/C to produce **A**. **B** reacts with O_3 , Zn/H_2O to produce **C**. **C** reacts with Ag_2O to produce **D**. **D** reacts with I_2 , OH^- to produce **E**. **E** under heating conditions yields the final product CC1CCC(CC1)=O.

Regarding the organic reaction in the above image, it is known that:

1. The molecular formula of **A** is C_9H_{18} .
2. The molecular formula of **C** is $C_9H_{16}O_2$.
3. The molecular formula of **E** is $C_8H_{14}O_4$.

Which of the following descriptions about **A** is correct:

- A.** All other options are incorrect
- B.** **A** must be chiral
- C.** **A** contains a six-membered ring
- D.** **A** containing a five-membered ring

E. A containing a four-membered ring

F. A Contains a triple bond

G. A contains the bonding relationship $\text{H}_3\text{C} - \text{C} - \text{C} - \text{CH}_3$

H. A contains an ethyl group.

I. A does not contain tertiary carbons

J. A containing isopropyl

K. A Containing terminal alkene

Answer

Correct Answer: **A**

Detailed Explanation

This question examines the ability of retrosynthetic analysis.

The final product is 4-methylcyclohexanone, which is generated by heating, and the molecular formula of **E** is $C_8H_{14}O_4$, indicating the loss of one molecule of CO_2 and two molecules of H_2O . Therefore, the first reaction should be decarboxylation, and **E** contains a carboxyl group;

CHECKPOINT

1 PTS

Heating loses one molecule of CO_2 and two molecules of H_2O

CHECKPOINT

1 PTS

E contains a carboxyl group

The loss of water produces a ketone carbonyl group, and the final product does not have a carbon-carbon double bond. The six-membered ring is entirely composed of carbon atoms, indicating that the loss of one molecule of water is only the dehydration of a hydrated ketal;

CHECKPOINT

1 PTS

The loss of one molecule of water is only the dehydration of a hydrated ketal

The degree of unsaturation of **E** is calculated to be 2, and it contains a carboxyl group. Therefore, **E** either has a ring or a double bond; if **E** has a ring, then there is no double bond, and the two oxygen atoms other than the carboxyl group can only exist in the form of alcohols; but if there are alcohols, the final product cannot be without a carbon-carbon double bond, so **E** must not have a ring, and there must be another carbonyl group besides the carboxyl group;

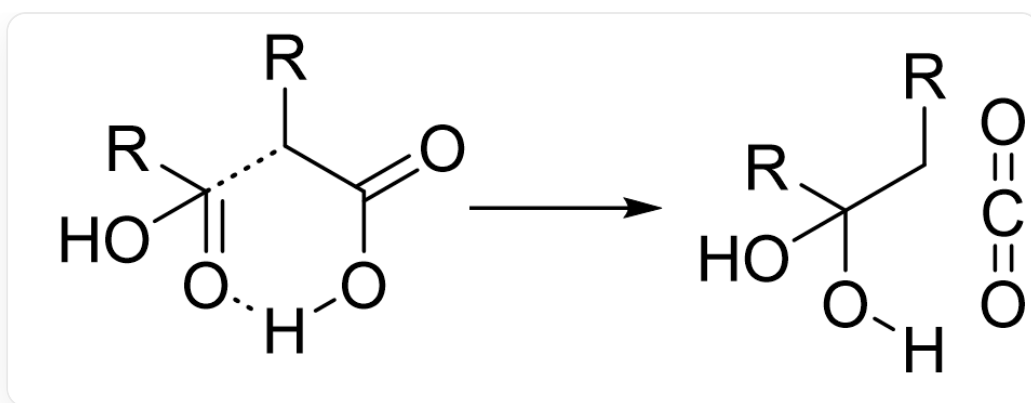
CHECKPOINT

1 PTS

E must not have a ring, and there must be another carbonyl group besides the carboxyl group

Combined with the fact that **E** loses one molecule of water in addition to the dehydration of a hydrated ketal, and the molecular formula of **E** contains four oxygen atoms, consider that **E** contains two carboxyl groups;

E must not have a ring, but the final product is a six-membered ring, so the newly formed carbon-carbon single bond on the ring can only be formed through a decarboxylation reaction; then at this time, we can think of the six-membered ring transition state decarboxylation process, as shown in the figure below:



This figure is a schematic diagram of the six-membered ring transition state decarboxylation; the left side is the state before decarboxylation, where two terminal carboxyl groups form a six-membered ring through the C=O1 of one carboxyl group and the C1-C2-O2-H2 of the other carboxyl group. C1 represents the α position of the carboxyl group, and C2 represents the carbon atom of the carboxyl group itself; a dashed line is drawn between C and C1, and a dashed line is drawn between H2 and O1. This transition state undergoes a reaction to generate the product on the right side of the figure, where C and C1 are connected, the single bond between C1 and C2 becomes a double bond, O2-H2 breaks, and H2 forms a bond with O1 of the other carboxyl group, forming an O=C=O structure and [H]OC(C[R])(O)[R]. In the figure, R represents other groups connected to the carboxyl group.

CHECKPOINT

1 PTS

The newly formed carbon-carbon single bond on the ring can only be formed through a decarboxylation reaction

CHECKPOINT

1 PTS

The decarboxylation reaction goes through a six-membered ring transition state, eventually yielding a hydrated ketal

This process removes one molecule of water and one molecule of carbon dioxide, generating a six-membered ring and eventually yielding a hydrated ketal, which is consistent with the previous deduction, so **E** contains two

terminal carboxyl groups. According to the position of the methyl group in the six-membered ring, the structure of **E** is deduced to be CC(CCC(O)=O)CCC(O)=O.

CHECKPOINT

1 PTS

E contains two terminal carboxyl groups

CHECKPOINT

1 PTS

The structure of **E** is CC(CCC(O)=O)CCC(O)=O

E is generated by the oxidation of **D** with iodine element under basic conditions, and the molecular formula of **C** has 1 more carbon atom than **E**, which is easily thought of as the oxidation process of methyl ketone. Methyl ketone undergoes a haloform reaction to generate carboxylic acid. One carboxylic acid in the structure of **E** is produced by the oxidation of methyl ketone, so the structure of **D** is CC(CCC(C)=O)CCC(O)=O.

CHECKPOINT

1 PTS

One carboxylic acid in the structure of **E** is produced by the oxidation of methyl ketone via iodoform reaction

CHECKPOINT

1 PTS

The structure of **D** is CC(CCC(C)=O)CCC(O)=O

At this time, the information may not be able to deduce, look back and observe other reactions: the reaction of **B** generating **C** is ozonolysis, which cleaves the double bond to generate two carbonyl groups; therefore, **B** contains an alkene.

The reaction from **B** to **A** is a hydrogenation reaction, **B** contains an alkene, so the chemical formula of **B** can be deduced to be C_9H_{16} ; thus, the degree of unsaturation of **B** is 2, and there is an alkene, indicating that **B** contains a ring; the chemical formula of **B** generating **C** only has two more oxygen atoms, so it is judged that **B** has an alkene on the ring, and ozonolysis generates two carbonyl groups.

CHECKPOINT

1 PTS

B has an alkene on the ring, and ozonolysis generates two carbonyl groups

The condition for **C** to generate **D** is silver oxide, which is a typical reaction of aldehyde being oxidized to carboxylic acid; therefore, the structure of **C** is CC(CCC(C)=O)CCC=O.

CHECKPOINT

1 PTS

C generating **D** aldehyde is oxidized to carboxylic acid

CHECKPOINT

1 PTS

The structure of **C** is CC(CCC(C)=O)CCC=O

According to the ozonolysis reaction, the structure of **B** can be deduced to be CC1CCC=C(C)CC1.

CHECKPOINT

1 PTS

The structure of **B** is CC1CCC=C(C)CC1

The reaction from **B** to **A** is a hydrogenation reaction, and the structure of **A** is CC1CCCC(C)CC1.

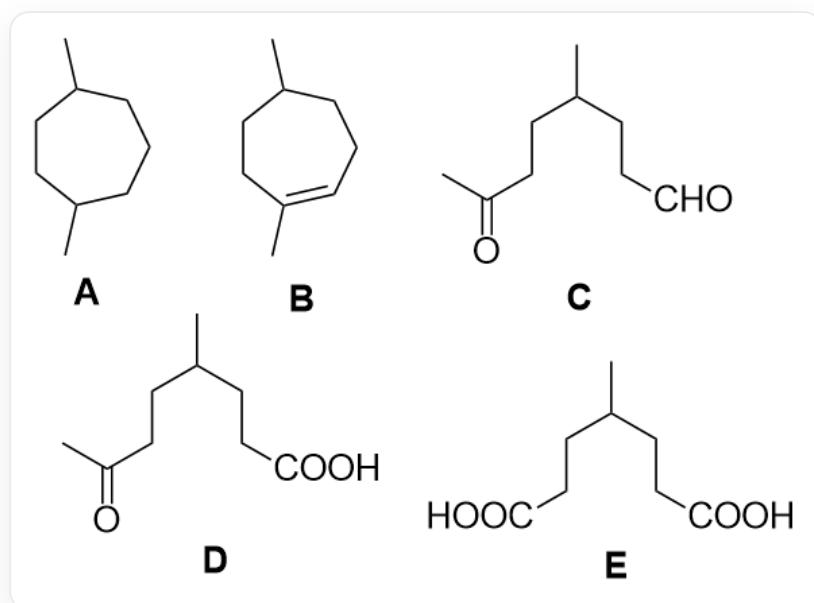
CHECKPOINT

1 PTS

The structure of **A** is CC1CCCC(C)CC1

According to the structure of **A**, options B-I are all incorrect.

In summary, option A is correct.



This figure shows the structures of the unknown substances **A** — **E** involved in this question. The structure of **A** is CC1CCCC(C)CC1; the structure of **B** is CC1CCC=C(C)CC1; the structure of **C** is CC(CCC(C)=O)CCC=O; the structure of **D** is CC(CCC(C)=O)CCC(O)=O; the structure of **E** is CC(CCC(O)=O)CCC(O)=O.